

STUDENT RESULT ANALYSIS

An Exploratory Data Analysis Report

Analyzing the factors that influence Math, Reading, and Writing performance among 30,641 students

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Tools used: Python · pandas · matplotlib · seaborn

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1. Introduction

Academic performance is shaped by far more than classroom teaching alone — a student's home environment, habits, and preparation all play a role. This report presents an exploratory data analysis (EDA) of a dataset of 30,641 students, examining how factors such as parental education, test preparation, sports activity, family size, and demographic background relate to performance in Math, Reading, and Writing.

The goal of this analysis is to identify which factors have a measurable impact on student scores and which do not, so that any future intervention or support strategy can be focused on what actually moves the needle.

2. Dataset Overview

The dataset contains the following structure:

Attribute	Detail
Total Records	30,641 students
Total Columns	14
Score Columns	MathScore, ReadingScore, WritingScore
Demographic Columns	Gender, EthnicGroup, ParentEduc, ParentMaritalStatus
Lifestyle Columns	PracticeSport, IsFirstChild, NrSiblings, TransportMeans, WklyStudyHours
Academic Prep Columns	LunchType, TestPrep

Several columns contained missing values, most notably TransportMeans (3,134 missing), ParentEduc (1,845 missing), and EthnicGroup (1,840 missing). These were reviewed during the cleaning stage below.

3. Data Cleaning

Before analysis, the following cleaning steps were applied:

- Removed the redundant index column ("Unnamed: 0") carried over from the CSV export.
- Standardized inconsistent category labels in WklyStudyHours (e.g. "5-10 Oct" corrected to "5-10").
- Checked all columns for null values using `df.isnull().sum()` prior to any grouping or visualization.

These steps ensured that grouping operations (e.g. by ParentEduc or TestPrep) reflected accurate, consistently labeled categories.

4. Gender Distribution

This chart shows the number of male versus female students in the dataset.

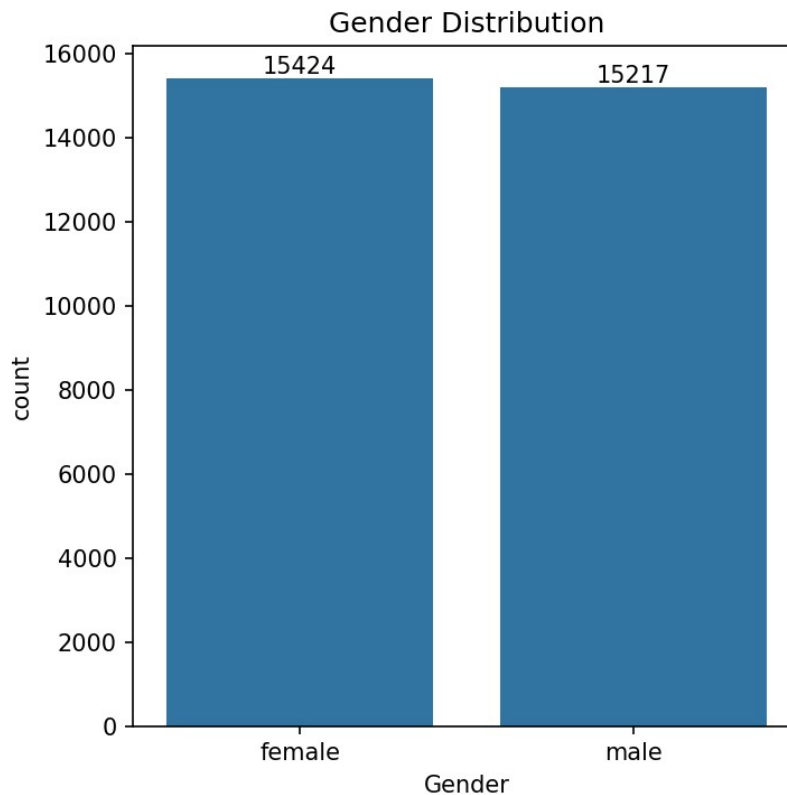


Figure 1: Gender distribution across the student population

- Female students slightly outnumber male students: 15,424 female (50.3%) vs. 15,217 male (49.7%).

Takeaway: The dataset is well balanced by gender, so any gender-based comparison of scores in this dataset can be considered statistically reliable, with neither group being too small to trust.

5. Parent Education vs. Student Score

This heatmap shows the average Math, Reading, and Writing scores grouped by parental education level, ordered from highest to lowest qualification.

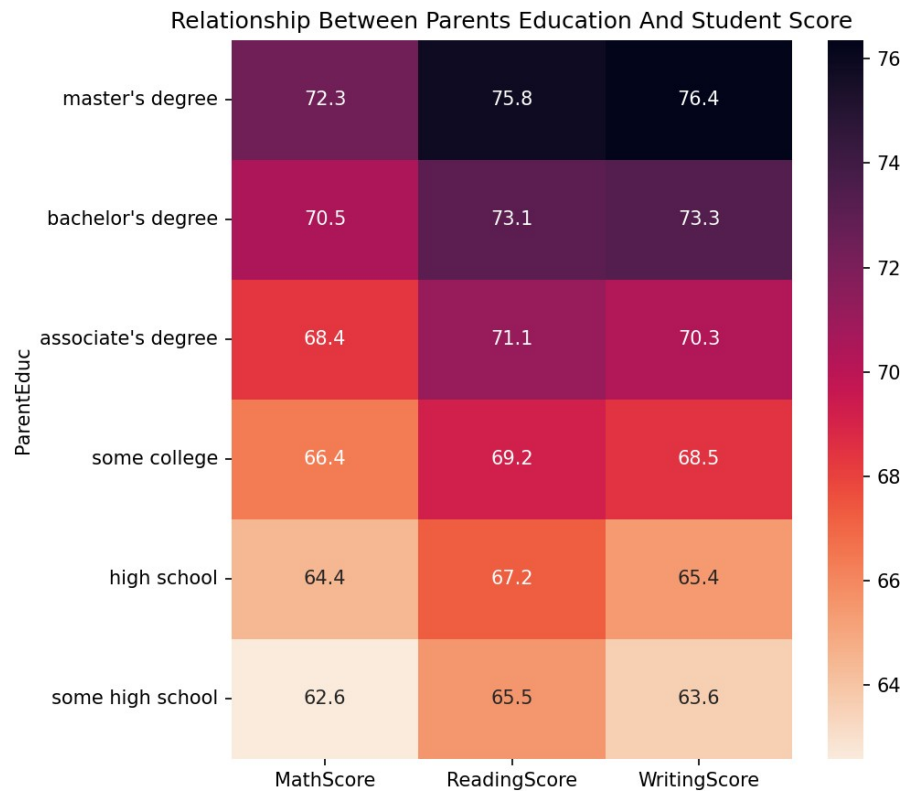


Figure 2: Average scores by parent education level

- Students whose parents hold a master's degree average the highest scores: Math 72.3, Reading 75.8, Writing 76.4.
- Students whose parents have only some high school education average the lowest: Math 62.6, Reading 65.5, Writing 63.6.
- The gap is largest in Writing (~12.7 points) and Reading (~10.3 points), and slightly smaller in Math (~9.8 points).

Takeaway: Parent education level shows a strong, consistent, and steadily increasing relationship with student performance across all three subjects — making it one of the most influential factors identified in this analysis.

6. Sports Practice vs. Student Score

This heatmap compares average scores across students who practice sports regularly, sometimes, or never.

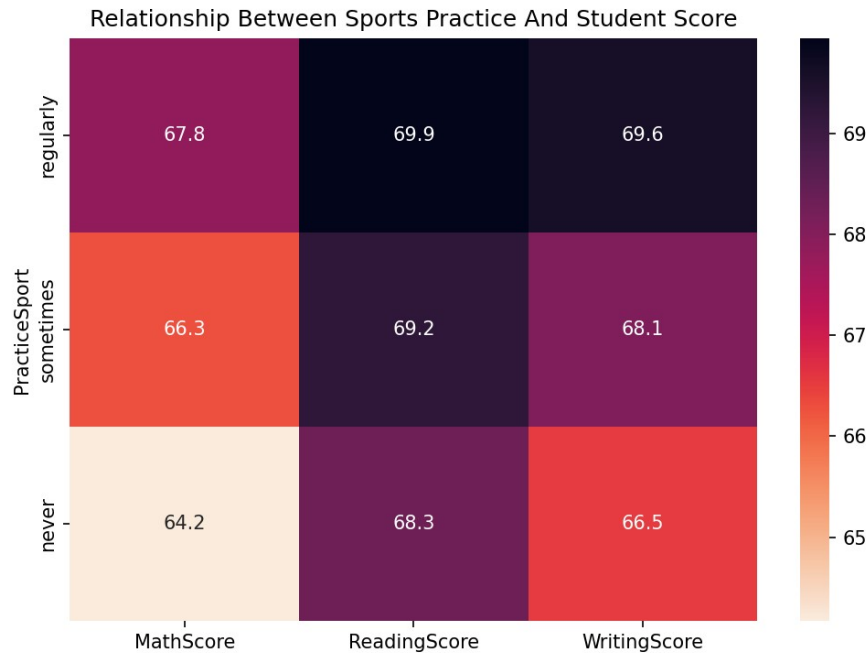


Figure 3: Average scores by sports participation frequency

- Students who practice sports regularly score highest: Math 67.8, Reading 69.9, Writing 69.6.
- Students who never play sports score lowest: Math 64.2, Reading 68.3, Writing 66.5.
- The gap between “regularly” and “never” groups is moderate (~2–3.6 points), smaller than the Parent Education effect.

Takeaway: Regular physical activity is associated with somewhat better academic performance, possibly linked to discipline, routine, or general well-being. This reflects a correlation, not proof of causation, but is a meaningful pattern worth noting.

7. Number of Siblings vs. Student Score

This heatmap shows average Math, Reading, and Writing scores for students grouped by number of siblings (0–7).

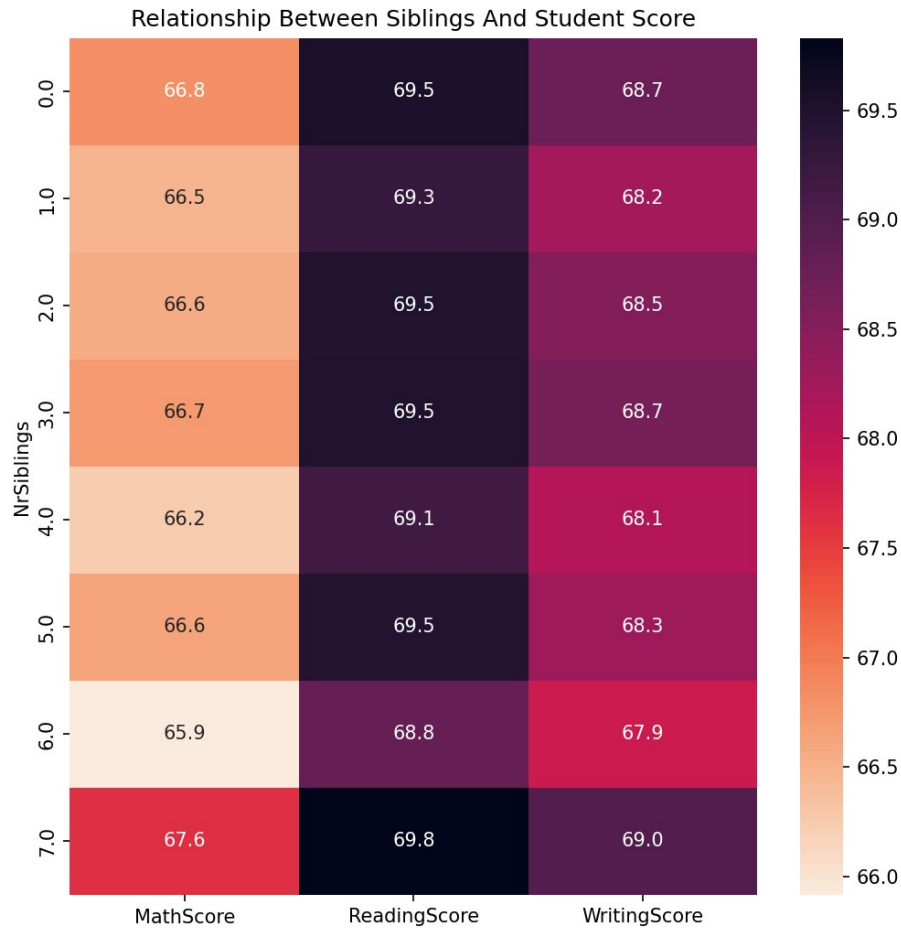


Figure 4: Average scores by number of siblings

- Scores remain remarkably stable across all sibling counts: Math 65.9–67.6, Reading 68.8–69.8, Writing 67.9–69.0.
- No consistent upward or downward trend is visible as sibling count increases.

Takeaway: Number of siblings has no meaningful impact on academic performance — scores stay essentially flat regardless of family size. This is a useful negative finding, ruling the factor out and helping focus attention on variables that do drive performance, such as Parent Education or Test Prep.

8. Outlier Detection

Boxplots were used to examine the distribution of each score column and the NrSiblings column, and to flag unusual values.

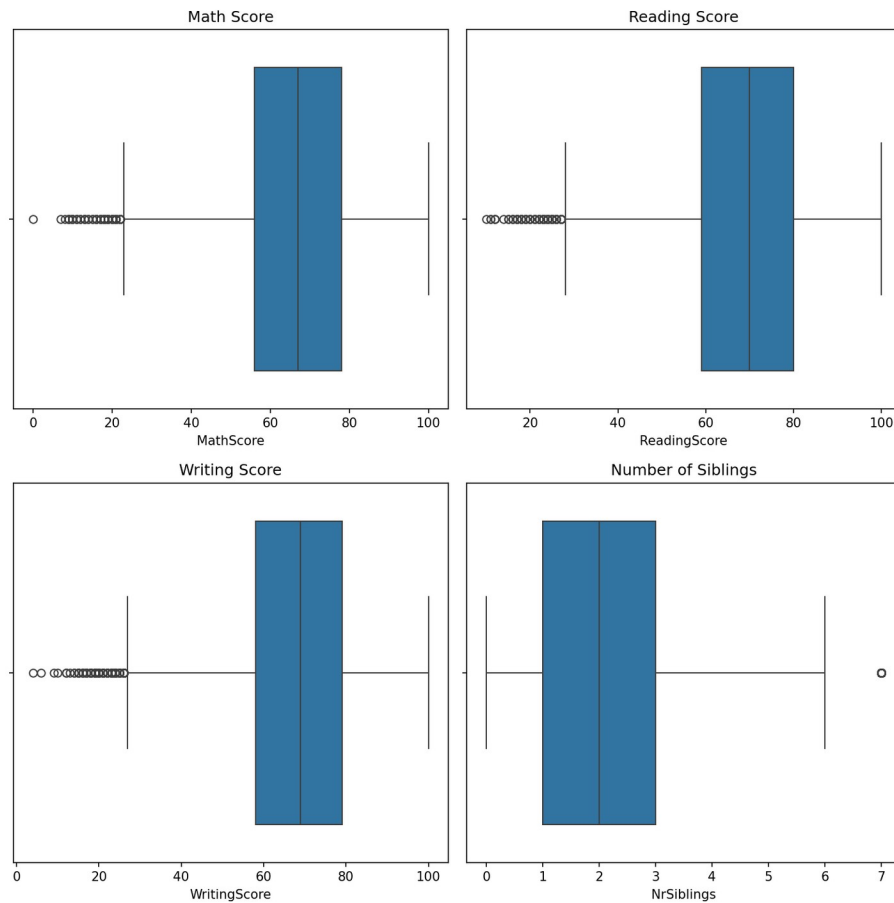


Figure 5: Boxplot distributions for Math, Reading, Writing scores and Number of Siblings

- MathScore, ReadingScore, and WritingScore each show a similar pattern: most students score between 55 and 80, with a small cluster of low-score outliers below roughly 25.
- No high-end outliers were found in any subject — top scores near 100 fall within the expected range.
- NrSiblings is fairly evenly spread between 0 and 3, with one outlier at 7 siblings, a value far higher than the rest of the dataset.

Takeaway: A small group of students consistently scored unusually low across all three subjects. These cases are worth flagging separately for review, as they could reflect genuine low performance or data entry issues. The single sibling-count outlier appears to be a rare case and does not affect the overall trend.

9. Ethnic Group Distribution

This chart shows how the student population is distributed across five ethnic groups.

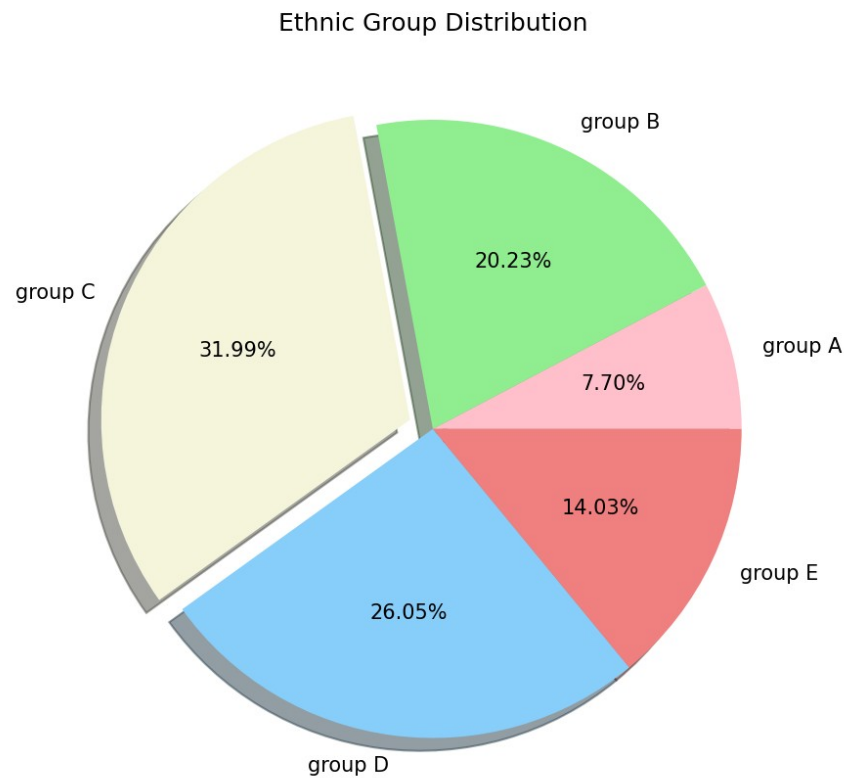


Figure 6: Distribution of students across ethnic groups

- Group C is the largest, at 9,212 students (31.99% of students with a recorded ethnic group).
- Group D follows with 7,503 students (26.05%), and Group B with 5,826 students (20.23%).
- Group E accounts for 4,041 students (14.03%), and Group A is the smallest at 2,219 students (7.70%).

Note: 1,840 students had no recorded EthnicGroup value; percentages above are calculated from the 28,801 students with a valid entry, matching the chart.

Takeaway: The sample is skewed toward Groups C and D, who together represent more than half of the dataset. Group A's smaller sample size means conclusions drawn specifically about that group should be treated with some caution.

10. Correlation Between Scores

This heatmap shows the pairwise correlation between MathScore, ReadingScore, and WritingScore.

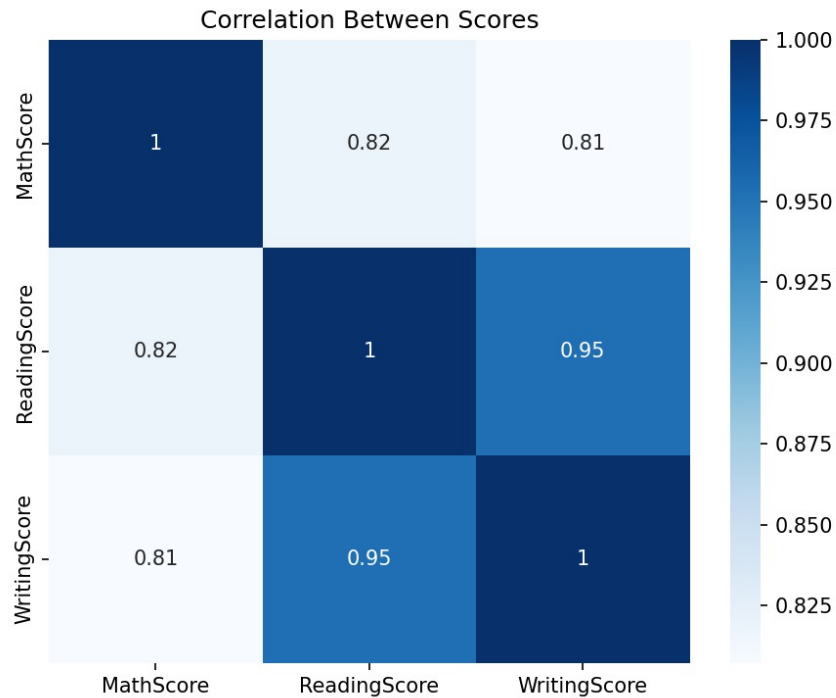


Figure 7: Correlation matrix of the three subject scores

- ReadingScore and WritingScore show the strongest relationship (0.95) — students who read well almost always write well, consistent with both being language-based skills.
- MathScore correlates somewhat less strongly with Reading (0.82) and Writing (0.81).

Takeaway: Reading and Writing skills move together almost in lockstep, while Math performance, though still related, reflects a comparatively distinct skill set.

11. Test Prep Completion vs. Student Score

This heatmap compares average scores between students who completed test preparation and those who did not.

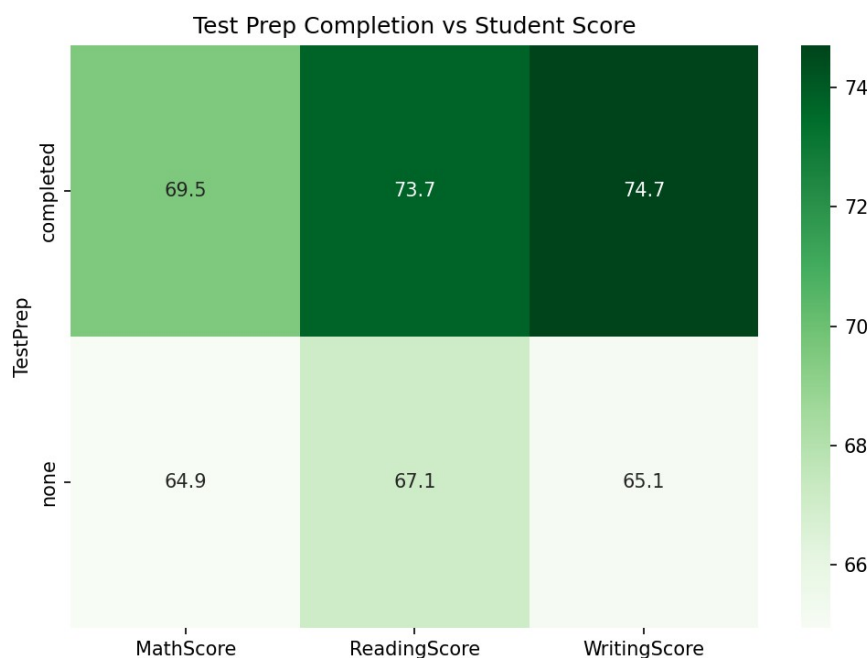


Figure 8: Average scores by test prep completion status

- Students who completed test prep scored higher across all subjects: Math 69.5, Reading 73.7, Writing 74.7.
- Students with no test prep averaged lower: Math 64.9, Reading 67.1, Writing 65.1.
- The gap is largest in Writing (+9.6 points), followed by Reading (+6.7 points) and Math (+4.6 points).

Takeaway: Completing test prep is associated with a meaningful score boost across all subjects, with the strongest effect on Writing. This makes Test Prep one of the most impactful factors identified in this analysis, on par with Parent Education.

12. Test Prep Completion Count

This chart shows how many students completed test prep versus how many did not.

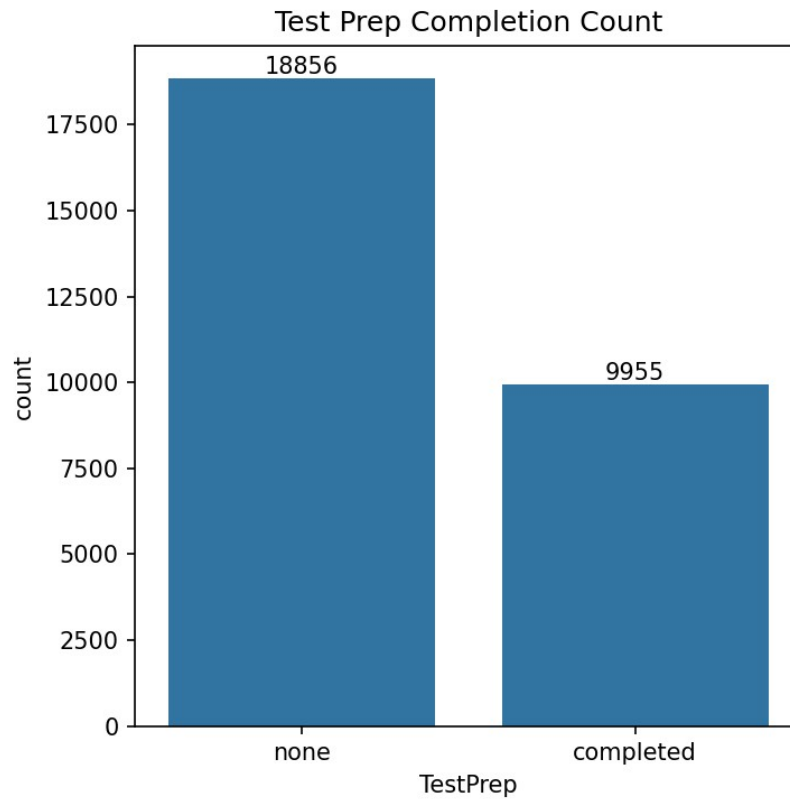


Figure 9: Number of students who completed test prep vs. did not

- 18,856 students (~65.4%) did not complete test prep.
- Only 9,955 students (~34.6%) completed test prep.

Takeaway: Nearly two-thirds of students skipped test prep entirely. Even though this smaller group still significantly outperformed the rest, the finding also points to an opportunity: wider test prep adoption could meaningfully raise average scores across the student population.

13. Final Conclusions

This analysis examined 30,641 students across multiple factors to identify what actually influences academic performance. Overall average scores were Math 66.6, Reading 69.4, and Writing 68.4 — Reading and Writing students typically scoring a few points higher than Math.

Factors that strongly influence scores

- **Parent Education** — the single biggest factor identified. Scores rise steadily from parents with “some high school” education (lowest) to parents with a master's degree (highest), with up to a 12.7-point gap in Writing.
- **Test Prep Completion** — students who completed prep scored 4.6 to 9.6 points higher across all subjects, with the strongest boost in Writing. Only 34.6% of students completed prep, suggesting room for improvement if more students took it up.
- **Sports Practice** — students who play sports regularly outperform those who never do, though the gap is smaller than for Parent Education or Test Prep.

Factors with little to no effect

- **Number of Siblings** — scores stayed essentially flat (within 1–2 points) regardless of family size, ruling this out as a meaningful driver of performance.

Other observations

- Gender is nearly balanced (50.3% female vs. 49.7% male), so gender-based comparisons in this dataset are statistically reliable.
- Reading and Writing scores are tightly correlated (0.95) — students strong in one are almost always strong in the other — while Math is comparatively more independent (0.81–0.82 correlation with the others).
- Ethnic Group distribution is uneven: Groups C (31.99%) and D (26.05%) together make up more than half the student population, while Group A is the smallest (7.70%).
- A small cluster of students scored consistently low across all three subjects, flagged as outliers worth a closer look — possibly genuine low performers or data entry issues.

Bottom line: Parent education and test prep completion are the strongest predictors of student performance, while family size has no measurable impact. If the goal is improving outcomes, increasing test prep accessibility — since nearly two-thirds of students currently skip it — stands out as the most actionable lever for raising scores broadly.